



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to RASIPURAM LORRY OWNERS ASSOCIATION

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 40.41 km
Estimated Duration: 1.1 hours

Adjusted Duration (Heavy Vehicle): 1.4 hours

Start: (11.4381, 77.8734)

End: (11.471761, 78.157623)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route spans approximately 40.41 kilometers and passes through key locations, starting from CVQF+23W, Sangagiri, Tamil Nadu, moving through CXV6+47Q, Namakkal, and FXCW+8C7, Mallasamudram West, before reaching the TNSTC Rasipuram Bus Depot. This path follows a series of state and national highways that connect these important points.

2. Typical Weather Conditions and Potential Weather-Related Hazards

The region generally experiences a tropical climate with hot summers, moderate monsoons, and mild winters. During the monsoon season (June to September), heavy rainfall may cause flooding or waterlogging in some low-lying stretches, potentially increasing the risk of accidents. Visibility can also become an issue during heavy rains.

3. Analysis of Traffic Patterns

- **Peak Hours:** Traffic is typically heaviest from 8:00 AM to 10:00 AM and from 5:00 PM to 7:00 PM due to local commuting.
- **Congestion-Prone Areas:** Key congestion points include entry and exit areas of towns like Namakkal, and junctions around busier local markets and intersections.

4. Assessment of Road Quality and Infrastructure

- **State Highways:** Generally well-maintained, but occasional patchy sections may exist due to ongoing repairs.

- **Local Roads:** Road quality may vary, with certain stretches experiencing wear and tear, particularly after the monsoon season.

5. Suggestions for Alternative Routes for Emergencies

In the event of an emergency, such as roadblocks or severe congestion, consider the following alternative:

- Divert through neighboring towns using the Namakkal - Salem route, which is often less congested and offers well-maintained road infrastructure.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- Strict restrictions exist on the timing of hazardous material transport, typically prohibited during peak hours to avoid accidents.
- Designated routes may be compulsory, especially for transporting flammable or chemical hazardous materials.

7. Overview of Historical Incidents

- There have been occasional incidents of heavy vehicle accidents, mainly attributed to human error and adverse weather conditions.
- Few recorded incidents involving hazardous materials, but existing ones emphasize the risks presented by inadequate safety measures.

8. Environmental Considerations and Sensitive Areas

- The route passes through some agricultural lands that are sensitive to chemical spills. Proper containment measures are crucial.
- Conservation-sensitive areas like parks or water bodies should be protected from exposure to hazardous materials.

9. Analysis of Communication Coverage

Cellular network coverage is generally reliable on major highways and near towns, but isolated dead zones may be present in rural stretches. Drivers should be equipped with GPS systems and emergency communication tools.

10. Estimated Emergency Response Times

- **Urban Areas:** Approximately 20-30 minutes, given access to local emergency services in towns like Namakkal and Rasipuram.
- **Rural Areas:** Up to 45 minutes, depending on the location and the severity of the emergency.

11. Overall Summary of Risk Assessment

Risks Identified:

- Moderate risks related to weather conditions during monsoon seasons.
- Potential road quality issues during repair periods or post-monsoon wear.
- Risks of congestion leading to delays, particularly during peak traffic hours.

Mitigation Measures:

- Schedule transport to avoid peak hours and adverse weather conditions.
- Regular vehicle maintenance and driver training for handling unexpected road conditions.
- Utilizing alternative routes when necessary and ensuring compliance with regulations for hazardous materials.

The route is generally safe but requires verifying weather conditions before departure, compliance with regulations, and readiness to adjust to dynamic traffic and environmental conditions.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	Medium	11.43822, 77.87348	30 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Blind Spot	Blind Spot	11.44861, 77.87429	10 KM/Hr
4	Turn	Medium	11.45478, 77.88925	30 KM/Hr
5	Turn	High	11.43675, 77.91693	15 KM/Hr
6	Turn	High	11.43684, 77.91706	15 KM/Hr
7	Turn	High	11.43926, 77.91705	15 KM/Hr
8	Turn	Medium	11.43930, 77.92282	30 KM/Hr
9	Turn	Medium	11.43967, 77.92345	30 KM/Hr
10	Turn	Medium	11.43979, 77.93009	30 KM/Hr
11	Turn	Medium	11.44030, 77.93066	30 KM/Hr
12	Turn	Medium	11.44067, 77.93203	30 KM/Hr
13	Turn	High	11.44088, 77.93223	15 KM/Hr
14	Turn	Medium	11.44127, 77.93218	30 KM/Hr
15	Turn	Medium	11.44163, 77.93244	30 KM/Hr
16	Turn	High	11.44243, 77.93244	15 KM/Hr
17	Turn	Medium	11.44285, 77.93440	30 KM/Hr
18	Turn	Medium	11.44093, 77.93821	30 KM/Hr
19	Turn	Medium	11.44550, 77.95329	30 KM/Hr
20	Turn	Medium	11.44401, 77.96239	30 KM/Hr
21	Turn	High	11.44387, 77.96283	15 KM/Hr
22	Turn	High	11.49063, 78.02680	15 KM/Hr
23	Turn	High	11.49135, 78.03252	15 KM/Hr
24	Turn	High	11.49183, 78.03257	15 KM/Hr
25	Turn	Medium	11.49355, 78.03703	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
26	Turn	Medium	11.49410, 78.03864	30 KM/Hr
27	Turn	Medium	11.49381, 78.04241	30 KM/Hr
28	Turn	Medium	11.49637, 78.04986	30 KM/Hr
29	Turn	High	11.49676, 78.05010	15 KM/Hr
30	Blind Spot	Blind Spot	11.49723, 78.05308	10 KM/Hr
31	Turn	High	11.49553, 78.05338	15 KM/Hr
32	Turn	Medium	11.49547, 78.05442	30 KM/Hr
33	Turn	High	11.49518, 78.05498	15 KM/Hr
34	Turn	Medium	11.49544, 78.05538	30 KM/Hr
35	Turn	Medium	11.49526, 78.05677	30 KM/Hr
36	Blind Spot	Blind Spot	11.49548, 78.05904	10 KM/Hr
37	Turn	High	11.49469, 78.05908	15 KM/Hr
38	Turn	High	11.49451, 78.05995	15 KM/Hr
39	Turn	High	11.49314, 78.05979	15 KM/Hr
40	Turn	High	11.49305, 78.05987	15 KM/Hr
41	Turn	Medium	11.49306, 78.06011	30 KM/Hr
42	Turn	Medium	11.49358, 78.06081	30 KM/Hr
43	Turn	High	11.49370, 78.06142	15 KM/Hr
44	Turn	High	11.49561, 78.06192	15 KM/Hr
45	Turn	High	11.49665, 78.06789	15 KM/Hr
46	Turn	High	11.49725, 78.06787	15 KM/Hr
47	Turn	Medium	11.49738, 78.06806	30 KM/Hr
48	Turn	High	11.49738, 78.06877	15 KM/Hr
49	Turn	High	11.49906, 78.06934	15 KM/Hr
50	Turn	Medium	11.49969, 78.07116	30 KM/Hr
51	Turn	High	11.50124, 78.07191	15 KM/Hr
52	Turn	Medium	11.50070, 78.07343	30 KM/Hr
53	Turn	Medium	11.50092, 78.07374	30 KM/Hr
54	Turn	Medium	11.50085, 78.07854	30 KM/Hr
55	Turn	Medium	11.50143, 78.07938	30 KM/Hr
56	Turn	High	11.50374, 78.08048	15 KM/Hr
57	Turn	High	11.50424, 78.08040	15 KM/Hr
58	Turn	Medium	11.51303, 78.08909	30 KM/Hr
59	Blind Spot	Blind Spot	11.51452, 78.08915	10 KM/Hr
60	Turn	High	11.51417, 78.09091	15 KM/Hr
61	Turn	Medium	11.51445, 78.09107	30 KM/Hr
62	Turn	Medium	11.51468, 78.09148	30 KM/Hr
63	Blind Spot	Blind Spot	11.51494, 78.09165	10 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
64	Turn	Medium	11.51485, 78.09173	30 KM/Hr
65	Turn	Medium	11.48976, 78.14718	30 KM/Hr
66	Turn	High	11.48720, 78.15384	15 KM/Hr
67	Turn	High	11.47225, 78.15722	15 KM/Hr
68	Turn	High	11.47212, 78.15769	15 KM/Hr
69	Turn	High	11.47177, 78.15767	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
3	hospital	Government Hospital, Vennandur	11.51701, 78.088631	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	college	Mahendra Engineering College	11.4770412, 78.0024214	30 km/h	Medium
1	school	Akv Schools	11.4850407, 78.0227236	30 km/h	Medium
2	school	Akv Schools	11.4854328, 78.0233558	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44861, 77.87429



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45478, 77.88925



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.43675, 77.91693



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.43684, 77.91706



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.43926, 77.91705



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43930, 77.92282



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43967, 77.92345



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43979, 77.93009



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44030, 77.93066



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44067, 77.93203



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44088, 77.93223



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44127, 77.93218



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44163, 77.93244



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44243, 77.93244



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44285, 77.93440



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44093, 77.93821



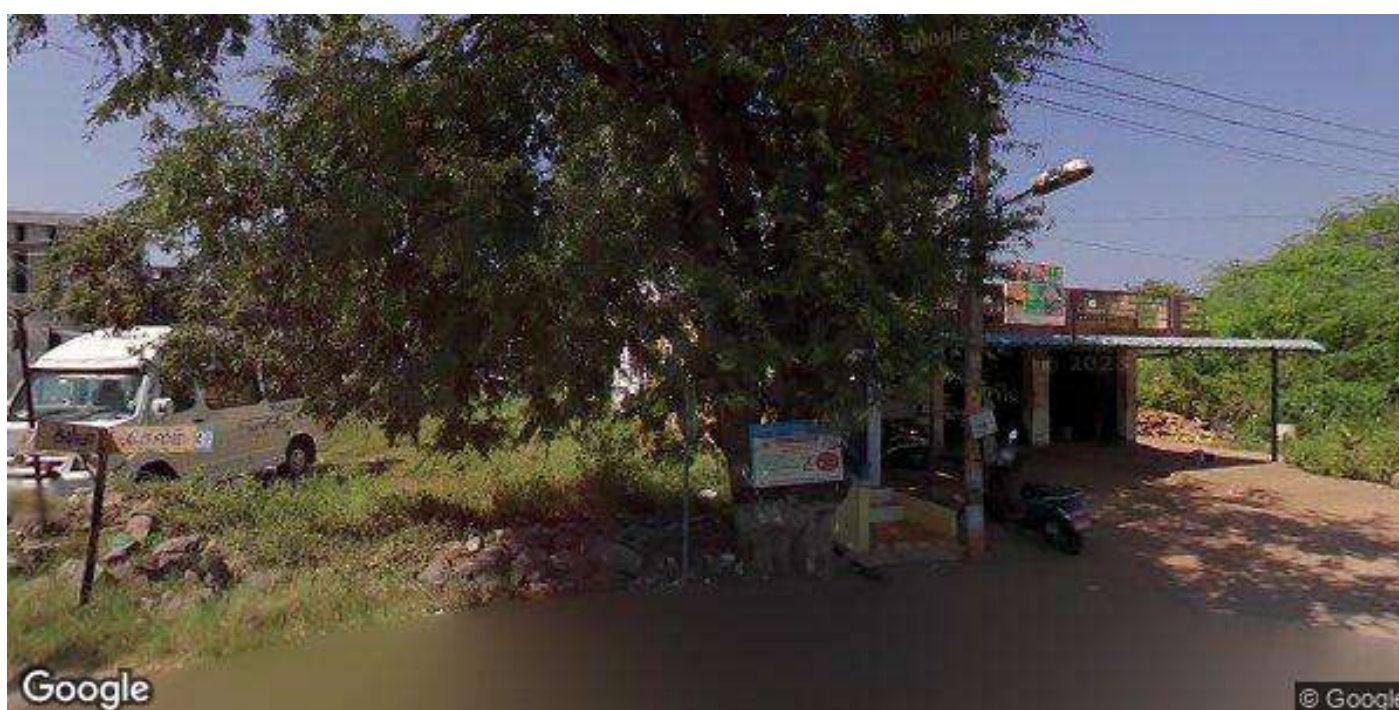
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Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44550, 77.95329



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44401, 77.96239



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44387, 77.96283



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49063, 78.02680



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49135, 78.03252



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49183, 78.03257



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49355, 78.03703



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49410, 78.03864



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49381, 78.04241



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49637, 78.04986



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49676, 78.05010



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49723, 78.05308



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49553, 78.05338



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49547, 78.05442



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49518, 78.05498



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49544, 78.05538



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49526, 78.05677



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49548, 78.05904



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49469, 78.05908



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49451, 78.05995



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49314, 78.05979



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49305, 78.05987



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49306, 78.06011



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49358, 78.06081



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49370, 78.06142



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49561, 78.06192



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49665, 78.06789



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49725, 78.06787



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49738, 78.06806



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49738, 78.06877



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49906, 78.06934



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49969, 78.07116



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.50124, 78.07191



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50070, 78.07343



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50092, 78.07374



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50085, 78.07854



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50143, 78.07938



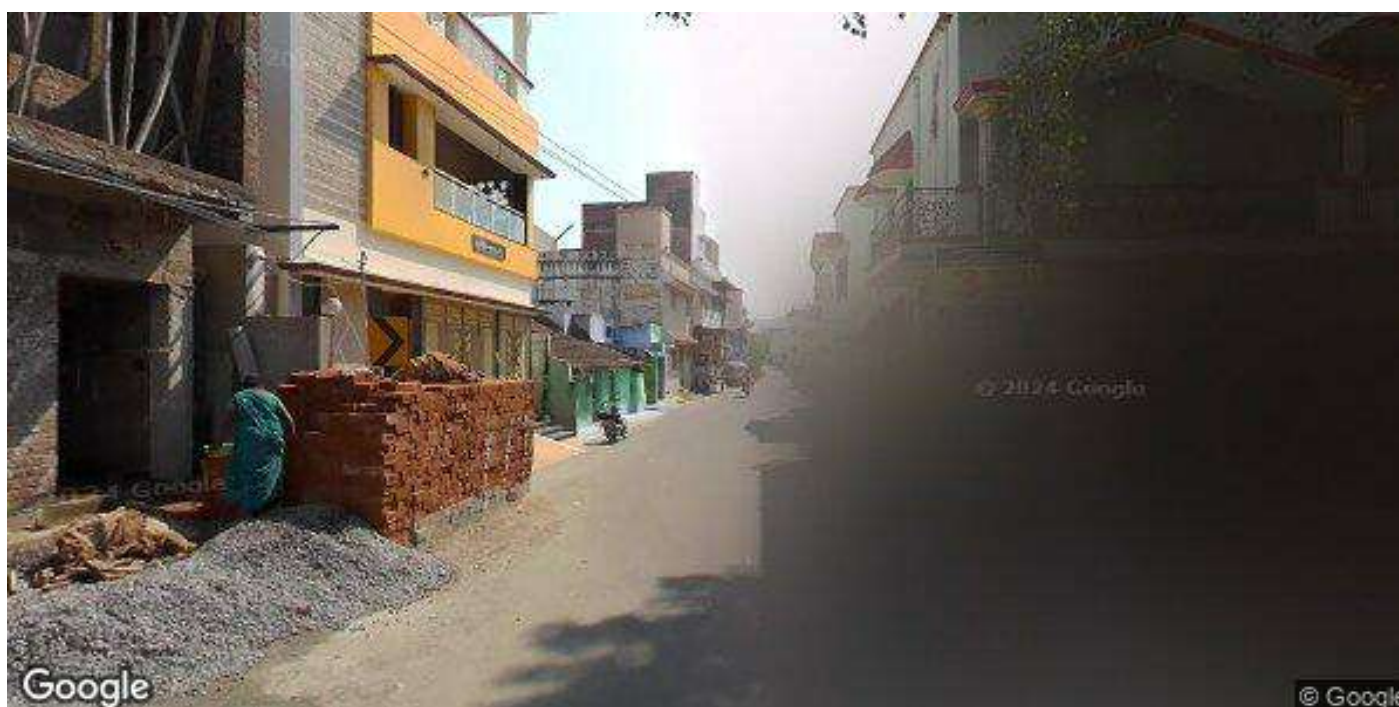
Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.50374, 78.08048



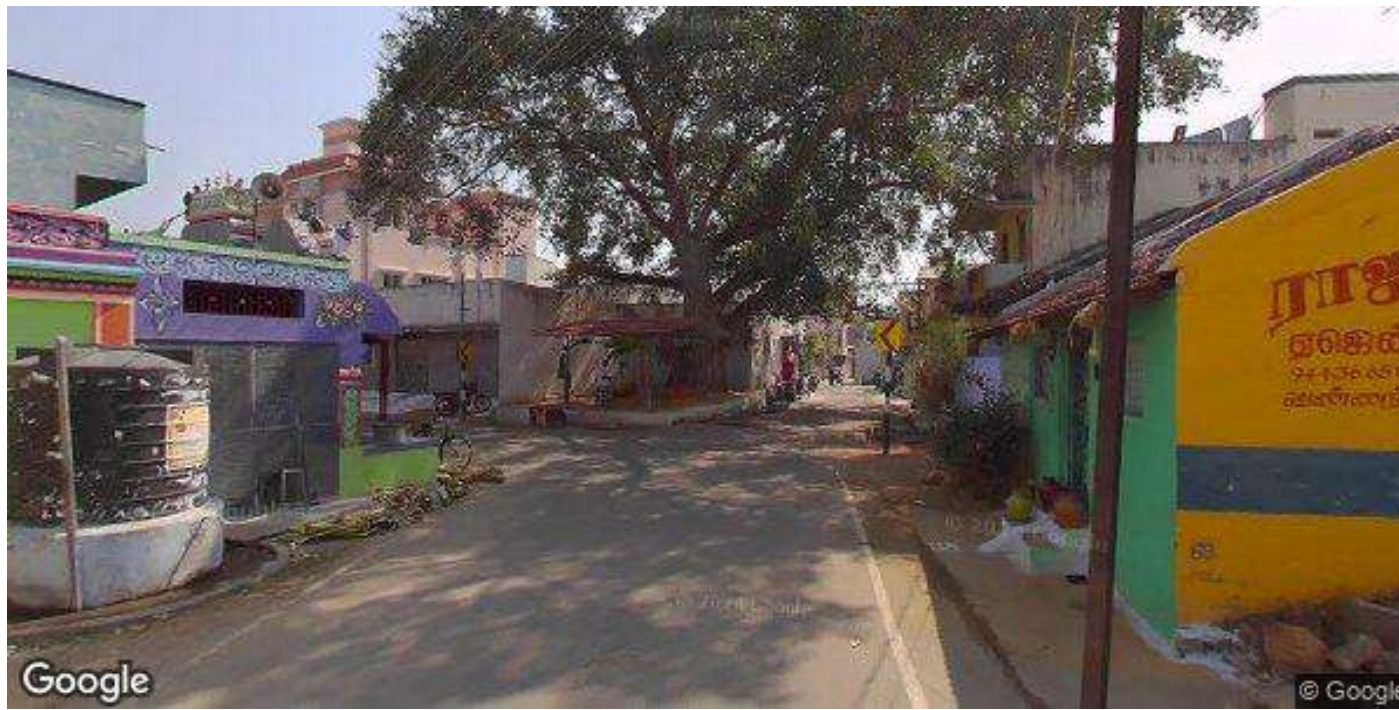
Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.50424, 78.08040



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.51303, 78.08909



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.51452, 78.08915



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.51417, 78.09091



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.51445, 78.09107



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.51468, 78.09148



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.51494, 78.09165



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.51485, 78.09173



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.48976, 78.14718



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.48720, 78.15384



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47225, 78.15722



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47212, 78.15769



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47177, 78.15767